What is in `embeddings.npz`?

Aviral Garg · Forgis screening task · 10 Oct 2026

Every number here comes out of `code/analyze.py` (numpy/scipy/torch, no sklearn, seed 0, about 4 minutes on CPU). Raw values are in `out/results.json`.

1. Identification

Given: `T` (2500, 512) f32, `I` (2500, 512) f32, `Y` (2500,) i64. No other metadata.

- `Y` is a class label: 31 distinct values in [1, 34], 31–33 absent ⇒ a subset of a 34-class dataset. 49–150 samples/class; majority class 6.0%.
- Not ℓ_2-normalised: $\|\texttt{T}\| = 9.06 \pm 0.86$, $\|\texttt{I}\| = 10.19 \pm 0.46$. Raw encoder output, taken before any normalisation layer.
- Two different encoders: per-dim variance correlation 0.0017, mean per-dim correlation 0.18, and only 4 of 512 dims reach $|r| > 0.5$. They share no basis.
- `T`'s inputs repeat exactly, `I`'s never do: `T` 2075/2500 unique (173 duplicate groups, max multiplicity 30), `I` 2500/2500. 98.8% of duplicate groups sit inside one class. Note 83% of `T` rows are still unique, so this is a pool with class-confined repeats rather than one string per class.
- Row order carries nothing: $\cos(\texttt{I}_i, \texttt{I}_{i+1}) = 0.6579$ against 0.6581 for a random pair, and the same at lags 2, 5, 10 and 50. `Y` has 2416 runs in 2500 rows.

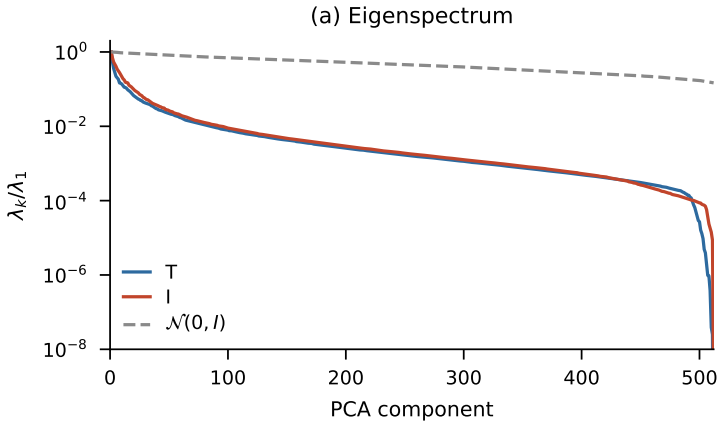
Hypothesis: a CLIP-style dual encoder, with `T`=text, `I`=image and `Y`=class.

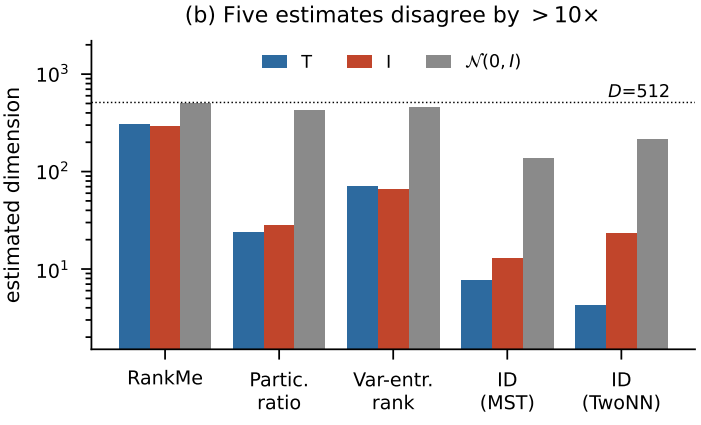
- For: D=512 is CLIP ViT-B/32's shared width for both towers; the vectors are unnormalised, as a dump taken before the ℓ_2 layer would be; two towers explain the disjoint bases; repeated captions explain `T`'s duplicates; and the mean-offset (§4) matches CLIP's documented gap.
- Against / open: paired similarity barely exceeds same-class similarity (§4), so the row pairing is class-level; and 34 classes on their own weren't enough for me to pin down the source dataset. I'm putting this forward as a hypothesis with its evidence, and nothing stronger.
- Decisive test, which needs the source data: encode known image/caption pairs with ViT-B/32 and compare norms, gap and spectrum. I have not measured reference CLIP norms myself.

It is not a HEPA dump. HEPA (Petersen et al., arXiv:2605.11130) emits a (h_t, h^*) pair per sample, so I tested for that. Three checks, all of them negative:

- D=512 ≠ HEPA's d=256 (their §3.1).
- No temporal structure, but HEPA's pairs index $(t, \Delta t)$ over episodes.
- HEPA's representations are ℓ_2-normalised onto the unit sphere (their appendix A.2, and table 7: "L1 on L2-normalised"), whereas $\|\texttt{T}\| = 9.06$, $\|\texttt{I}\| = 10.19$. I first used a SIGReg-distance test here; it does not discriminate, because HEPA applies SIGReg to the predictor output $\hat{h}$ and never to h_t or h^*, so a genuine dump need not look isotropic.

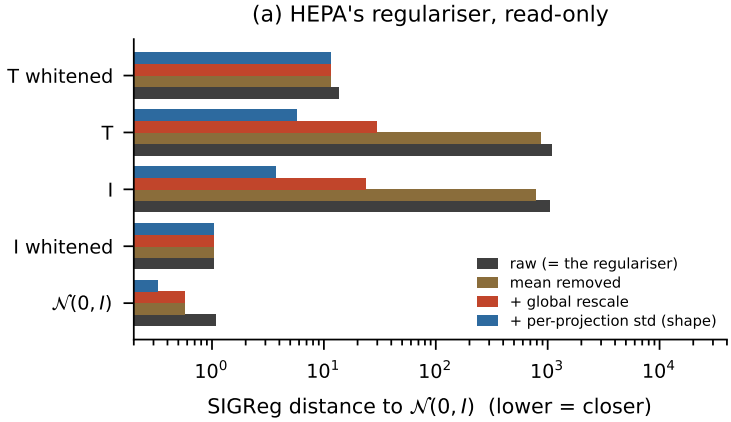
2. Geometry: every rank estimate disagrees

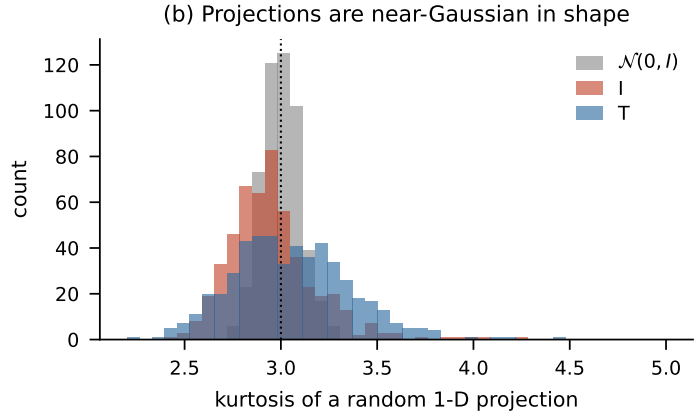




- Strongly anisotropic: 90% of variance needs 110 (`T`) / 93 (`I`) of 512 dims, after which the spectrum falls five orders of magnitude.
- Five standard estimates span > 10× on the same data (`T`/`I`): RankMe 309/292, participation ratio 24/28, variance-entropy rank 71/66, MST-ID 7.7/12.9, TwoNN-ID 4.3/23.2, and they disagree on the ordering: TwoNN puts `T`≪`I`, RankMe puts `T`>`I`.
- On an isotropic Gaussian control both ID estimators badly underestimate (136, 216 vs. a true 512), so at N=2500, D=512 absolute ID values are untrustworthy. I compare only within one estimator and name it every time.

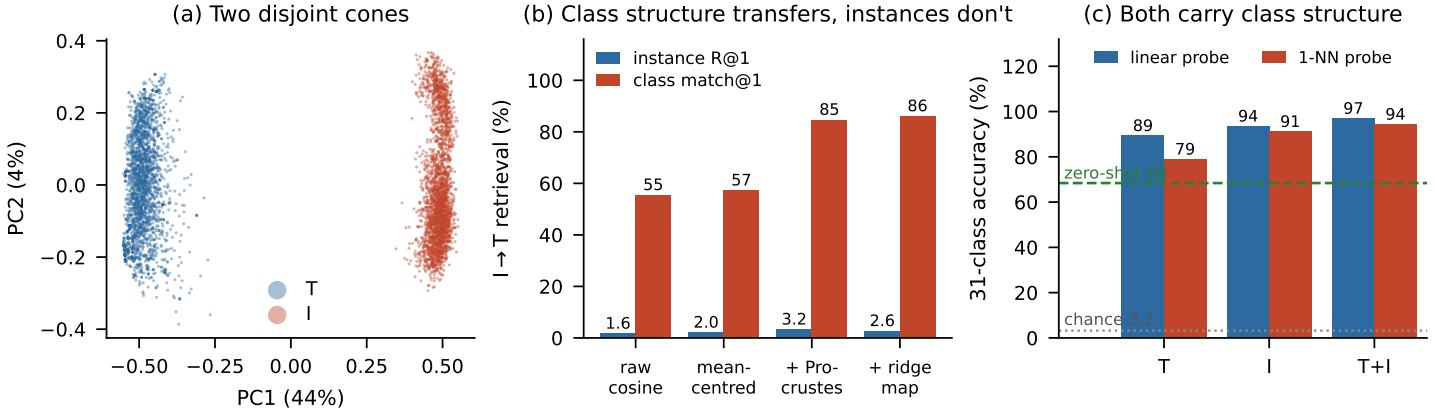
3. Isotropy, measured with HEPA's own regulariser





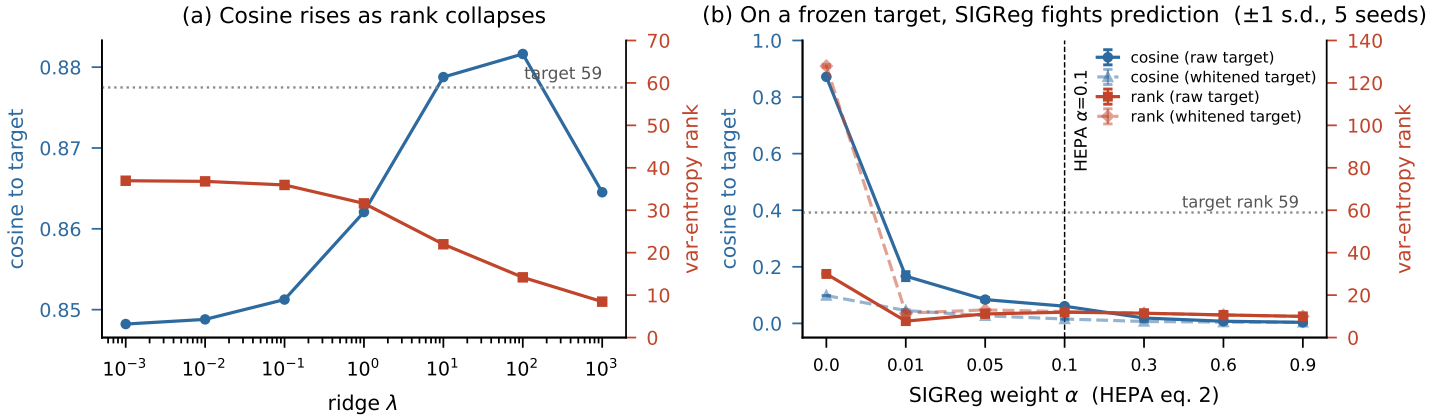
- raw is LeJEPA algorithm 1 verbatim (arXiv:2511.08544): random unit directions, no centring, no rescaling, Epps–Pulley with window $e^{-t^2/2}$, 17-knot trapezoid on $[-5, 5]$. Only that line is the regulariser; the next three strip one factor each to locate the departure (`T` / `I` / $\mathcal{N}(0, I)$):
  - raw 1091 / 1045 / 1.08 · mean removed 860 / 781 / 0.57 · + one global rescale 29.8 / 23.9 / 0.57 · + per-projection standardised 5.7 / 3.7 / 0.33.
- Almost all of the departure is scale and anisotropy; the shape is already near-Gaussian. Mean projection variance 0.044 / 0.070 where SIGReg wants 1.0; projection kurtosis 3.06 / 2.95 against 3.00. This drives §5.
- Whitening `T` makes it worse: shape-only 5.7→11.4, kurtosis 3.06→4.15, because whitening amplifies near-null directions into heavy tails. Whitening `I` behaves (1.04), so a blanket "whiten first" step isn't safe here.

4. The modality gap is a rotation for class structure, not for instances



- Protocol: centring, rotation and ridge map are fitted on the 2000 training rows; retrieval is queried with the 500 held-out rows against the full 2500-row gallery, so chance R@1 stays 0.04%. An earlier version fitted and scored on all 2500 rows and read 1.6% → 17.6% at instance level against the 3.2% below; class match moved far less (88.2% vs 84.6%), so the finding survives, narrowed.
- Disjoint clouds: $\cos(\bar{\mathtt{T}}, \bar{\mathtt{I}}) = 0.319$, $\|\bar{\mathtt{T}} - \bar{\mathtt{I}}\| = 0.97$ (normalised). PC1 alone separates them.
- Raw cosine barely sees the pairing: paired 0.278, same-class 0.272, random 0.222. The 0.006 margin says the correspondence is class-level.
- The rotation transfers at class level and barely at instance level (`I`→`T`, held-out): class match@1 55.4%→84.6%, but instance R@1 only 1.6%→3.2% (80× chance), R@5 7.2%→10.8%. Mean-centring alone gives 2.0% / 57.2%. That split is the finding: one orthogonal map recovers most of the class geometry across the gap and almost none of the instance correspondence, which is what the class-level pairing above predicts.
- Zero-shot (held-out `I` classified by train-fitted class-mean `T` prototypes): 68.4% → 89.0% after the same rotation; chance 3.2%.
- Not a pure rotation: an unconstrained ridge map edges class match (86.0%) and loses instance R@1 (2.6%), with singular values far from flat (mean 0.22, s.d. 0.34).
- Both carry class structure: linear probe 89.4 / 93.6 / 97.0% and 1-NN 79.0 / 91.2 / 94.4% (`T` / `I` / concat). The linear−kNN gap is +10.4 pts on `T` but +2.4 on `I`, so `T`'s class information is linearly decodable while being less metrically organised. Repeated captions would do exactly that.

5. Prediction in embedding space: cosine improves, nothing downstream does



- Setup: the obvious JEPA experiment here. Context `T`, target `I`, predict in representation space. Held-out metrics; MLP results are mean ± s.d. over 5 seeds.
- A ridge map looks better and better while collapsing. Over $\lambda : 10^{-3} \to 10^{2}$, cosine rises 0.848→0.882 and R^2 0.13→0.34, while the prediction's variance-entropy rank falls 37.0→14.2 against a target rank of 58.9. Had I reported cosine alone it would have hidden a 2.6× rank collapse.
- But rank does not predict the downstream number either. Over that same range held-out retrieval is flat (R@1 8.0, 7.8, 7.8, 8.2, 8.4, 7.8%) and only falls to 4.2% at λ=10^3. So cosine improves monotonically while buying nothing downstream, and rank tracks the collapse but not the task. Report all three.
- SIGReg on the predictor does not fix it here. 2-layer MLP under HEPA eq. (2), $\mathcal{L} = (1-\alpha)\|\hat{h} - h^*\|_1 + \alpha\mathcal{L}_{\text{SIGReg}}$: at α=0, cosine 0.8716±0.0004 / rank 30.0±0.1; at HEPA's own α=0.1, cosine 0.0609±0.0044 / rank 11.9±0.2. Every $\alpha \geq 0.01$ destroys prediction, with seed spread far below the effect.
- Part of it is a scale conflict HEPA does not have. HEPA trains "L1 on L2-normalised" representations (table 7), so both sides sit on the unit sphere; my frozen target has mean projection variance 0.070 while SIGReg wants 1.0. Control: against a whitened target (variance 1.000) that conflict is gone and prediction still collapses (cosine 0.0985±0.0004→0.0455±0.0034 at α=0.01; rank 128.0±0.7→11.5±0.7). So the remaining effect is about the target being frozen, not about its scale.
- What this does and does not say about HEPA. HEPA's target is never frozen: weight-shared and jointly trained (§3.1), and their appendix I.3 shows the method is robust to an EMA target instead (τ=0.99; joint wins 6/12, loses 4, ties 2, all within ±0.11 h-AUROC), choosing joint training "for its simplicity". Both leave the target free to move toward the Gaussian SIGReg wants, and even τ=0.99 is far more mobile than frozen. I measured the frozen-target regime, which neither setting in I.3 covers: there, no α bought both prediction and rank. That gap is where my proposal starts.

6. Caveats

- One 80/20 split throughout, seed 0: every fitted object in §4 and §5 is fitted on the 2000 training rows and scored on the 500 held out. I don't resample it, so the split is an untested source of variance. MLP results are 5 seeds; the ridge sweep is exact. §2–§3 are descriptive statistics of all 2500 rows.
- The CLIP identification is unconfirmed; the test is named in §1.
- The α sweep is one architecture at one budget (2-layer MLP, 200 epochs) on unnormalised targets. I'd stand behind the direction, that SIGReg and the prediction term conflict on a frozen target, not the crossover point. I have not run HEPA itself. Next: resample the split, and find where the rotation's residual 15.4% class error concentrates.